ZENVE INTELLIGENCE ENGINE

Enterprise Settings & Governance Suite — Full Architectural Specifications

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Scope: All 14 Settings Modules
PRODUCTION VERIFIED

Platform Settings & Enterprise Governance Architecture

Complete structural and operational reference guide covering all 14 Settings dashboards in the Zenve BI Platform. Enforces healthcare statutory licensing, organizational cost centers, clinical RBAC, automated approval hierarchies, delivery SLAs, and disaster recovery.

01. Company Settings
02. Business Units
03. Locations
04. Users Management
05. Roles & Permissions
06. Approval Workflows
07. Notifications
08. Dashboard Settings
09. Tax & GST Engine
10. Payment Gateways
11. Delivery & Logistics
12. API & Integrations
13. Zero-Trust Security
14. Backup & Recovery

Host Controller: ZenveSettingsDashboard (assets/settings-dashboard.js)

1. Company Settings

`#company-settings`

Centralized corporate legal identity, statutory registration, statutory drug licenses, and national fiscal parameters.

🏢 LEGAL ENTITY PROFILE

- **Legal Entity Name:** Registered corporate name in MCA registry
- **Consumer Brand:** Public brand identity ("Zenve")
- **CIN:** Corporate Identification Number
- **PAN:** Permanent Account Number
- **Registered Address:** HQ legal address for tax filing

💊 HEALTHCARE LICENSES

- **Drug License (Form 20B/21B):** Wholesale/retail pharmaceuticals
- **VCI Accreditation:** Veterinary Council of India telemedicine certificate
- **Bio-Medical Waste:** SPCB biomedical waste disposal compliance code

🌐 FISCAL LOCALIZATION

- **Operating Currency:** INR (₹) base currency
- **Financial Year:** April 1 – March 31 cycle
- **Timezone:** Asia/Kolkata (IST, UTC+05:30)
- **Database Key:** Synchronized to MySQL `company_settings`

2. Business Units

`#business-units`

Organizational cost centers, vertical heads, and P&L allocations across hospitals, digital tele-health, pharmacy, and lifestyle retail.

🏛 DIVISION ARCHITECTURE

- **BU-01 (Clinical Hospitals):** Specialty surgical centers and 24/7 ICU facilities
- **BU-02 (Digital Tele-Health):** Remote video consultations and home vet visits
- **BU-03 (Pharmacy & Diagnostics):** Central pathology labs and cold-chain pharmacy
- **BU-04 (Pet Nutrition & Lifestyle):** Premium diets, prescription nutrition, and accessories
- **BU-05 (B2B & Corporate Wellness):** Enterprise accounts and corporate pet care benefits

📊 OPERATIONAL METRICS TRACKED

- **Division Lead:** Appointed vertical leader / Chief Medical Officer
- **Cost Center Code:** SAP/Tally compatible GL identifier (e.g., `CC-HOSP-BLR`)
- **Staff Count:** Active headcount allocated to the division
- **Revenue Run-Rate:** Monthly recurring gross realization per unit
- **Dynamic Provisioning:** Modal to instantiate new operational divisions

3. Locations

`#locations`

Multi-tier physical node management spanning flagship veterinary hospitals, trauma clinics, and dark stores.

FACILITY NODE TYPES

- **Super-Specialty Hospital:** Full surgery theatres & ICU
- **Micro-Fulfillment Hub:** 60-min express dark store
- **Specialty Surgical Center:** Orthopedic/cardiac surgery
- **Central Cold-Chain Hub:** Bulk vaccine depot
- **Outpatient & Diagnostic Node:** Walk-in clinic & lab

NODE METADATA

- **Node ID:** System code (e.g., `LOC-01` to `LOC-07`)
- **Facility & Area:** Locality, road, and pin-code
- **City Coverage:** Bengaluru, Mumbai, NCR, Hyderabad, Chennai
- **Coverage Radius:** Operating boundary (5 km – 25 km)
- **Manager Contact:** Station manager name & dedicated phone

REAL-TIME CONTROLS

- **Live Search Bar:** Filter by facility name, city, or area
- **Operational Status:** Active, Maintenance, or Offline
- **New Node Creation:** Modal form with address & type mapping
- **Dispatch Routing:** Linked to 60-min delivery dispatcher

4. Users Management

`#users`

Personnel identity provisioning, professional designations, facility assignments, and Two-Factor Authentication (2FA) status.

PERSONNEL IDENTITY

- **User ID:** Unique identifier (`USR-101+`)
- **Full Name & Work Email:** Enterprise domain credential
- **Avatar Initials:** Color-coded visual badges
- **Account Status:** Active, Suspended, or Pending Invite

ROLE & UNIT BINDING

- **Designated Role:** CMO, Vet, Pharmacist, Dispatch Lead, Auditor
- **Business Unit:** Clinical Hospitals, Supply Chain, Finance
- **Primary Facility:** Assigned flagship hospital or dark store

SECURITY COMPLIANCE

- **MFA Policy:** Enforced / Enabled flag tracking
- **Invitation Dispatch:** Modal wizard generating secure invite links
- **Search Index:** Real-time filtering across name, email, and roles

5. Roles & Permissions (RBAC)

`#roles-permissions`

Fine-grained Role-Based Access Control protecting clinical patient confidentiality, narcotics, dispatch, and ledgers.

MODULE SCOPE	VIEW	CREATE	EDIT	DELETE	EXPORT CSV	APPROVE
Clinical Patient Records (EHR)	✓	✓	✓	×	✓	✓
Veterinary Prescriptions (Rx)	✓	✓	✓	×	✓	✓
Orders & 60-Min Delivery Command	✓	✓	✓	×	✓	✓
Cold-Chain Pharmacy & Batch Inventory	✓	✓	✓	×	✓	✓
Financial Ledgers & Tax Reports	✓	×	×	×	✓	✓
User Management & Security Access	✓	✓	✓	✓	✓	✓
System API & Webhook Configuration	✓	✓	✓	×	✓	✓

Configured Roles: Super Administrator (Root) · Clinical Director & CMO (Medical Governance) · Senior Veterinarian (Clinical Diagnosis) · Operations & Dispatch Lead (Fulfillment) · Head Pharmacist (Controlled Drug Compliance) · Financial Auditor (Reconciliation & Tax).
Persistence: Changes sync interactively via `#btn-save-permissions`, persisting to localStorage and MySQL `rbac_permissions`.

6. Approval Workflows & Multi-Tier Sign-Offs

`#approval-workflows`

Enforces the four-eyes dual authorization principle on clinical narcotics, large purchase orders, and commercial refunds.

📋 CONFIGURED GOVERNANCE RULES

- **WF-01 (High-Value Refunds):** Triggered on customer refunds > ₹5,000. Chain: Care Coordinator ➔ Ops Manager ➔ Finance Lead (SLA: 4 Hours).
- **WF-02 (Controlled Narcotics / Sched-X):** Triggered on controlled drug Rx. Chain: Consulting Vet ➔ CMO Dual Sign-Off (SLA: 30 Mins).
- **WF-03 (Purchase Orders):** Triggered on inventory PO > ₹1,00,000. Chain: Procurement Specialist ➔ CFO Approval (SLA: 12 Hours).

⚡ INTERACTIVE EXECUTION QUEUE

- **Pending Approval Cards:** Displays item title, requested amount, requester, transaction reference, and submission timestamp.
- **Immediate Action Buttons:** Interactive ✓ `Approve` and ✕ `Reject` handlers with automated toast feedback.
- **Zero-Pending State:** Renders green validation prompt when all queues are fully cleared.

7. Notification Settings & Automated Alerts

`#notification-settings`

Omnichannel communication dispatch connecting WhatsApp Business, Transactional SMS, Amazon SES, and Firebase Push.

📱 CONNECTED DISPATCH GATEWAYS

- **WhatsApp Business API (Gupshup):** Pre-approved templates for digital prescriptions and live delivery map tracking.
- **Transactional SMS (DLT Approved):** High-priority OTP authentication and emergency dispatch notifications.
- **Transactional Email (Amazon SES):** Tax invoices, clinical consultation summaries, and medical history export PDFs.
- **Mobile Push (Firebase FCM):** Customer app status updates, rider arrival alerts, and booster reminders.

⚡ AUTOMATED TRIGGER EVENTS

- **Order Confirmation:** Dispatched instantly upon successful payment gateway authorization.
- **60-Min Express Dispatch:** Alerts customer when dedicated rider leaves the micro-hub.
- **Doctor Consultation Reminders:** Automated alerts sent at T-24 Hours and T-1 Hour before veterinary appointments.
- **Critical Low Medicine Stock:** Alerts head pharmacist when inventory of essential medicines falls below 15 units.
- **Test Ping Handler:** Dedicated button to fire immediate diagnostic webhook ping.

8. Dashboard Settings & Workspace Preferences

`#dashboard-settings`

Personalization of portal startup landing, live WebSocket data refresh rates, and presentation confidentiality masking.

🖥️ WORKSPACE PREFERENCES

- **Default Launch Dashboard:** Configurable landing view: Executive Control Center, Orders & Operations, or Sales.
- **Live Data Polling Cadence:** Selectable polling frequency: Live WebSocket (~10s), 30 Seconds, 1 Minute, or 5 Minutes.

🔒 BOARDROOM PRIVACY & CONFIDENTIALITY

- **Boardroom Revenue Masking Mode:** Instant toggle that obfuscates all gross financials and confidential patient totals with ₹ •••••.
- **Use Case:** Designed for board meetings, public demonstrations, and projector sharing without leaking financial metrics.

9. Tax Settings & GST Governance

`#tax-settings`

Statutory GST rate schedules, HSN/SAC classification codes, and veterinary healthcare exemption management.

MASTER GST COMPLIANCE

- **Primary GSTIN Number:** 15-digit statutory GSTIN linked to registered corporate entity.
- **TCS (Tax Collected at Source):** Statutory 1% operator deduction for marketplace sellers.
- **Automated E-Invoicing:** Direct API threshold sync with the National Informatics Centre (NIC) portal.

HSN / SAC CODE SCHEDULES

- **Veterinary Clinical Consultations (SAC 999312):** 0% GST (Exempt healthcare service).
- **Vaccines & Life-Saving Biologics (HSN 3002):** 5% GST (2.5% CGST + 2.5% SGST / 5% IGST).
- **General Pharmaceutical Medicines (HSN 3004):** 12% GST (6% CGST + 6% SGST / 12% IGST).
- **Pet Nutrition & Accessories (HSN 2309):** 18% GST (9% CGST + 9% SGST / 18% IGST).

10. Payment Gateways & Payout Orchestration

`#payment-settings`

Multi-gateway routing, instant doctor disbursements, settlement accounts, and Cash-on-Delivery (COD) risk safeguards.

GATEWAY ROUTING

- **Primary Gateway (Razorpay):** UPI AutoPay, credit cards, debit cards, and net-banking.
- **Secondary Gateway (Cashfree):** Instant doctor consultation payouts & vendor disbursements.
- **Cross-Border (Stripe):** International pet owners & telemedicine consults.

SETTLEMENT ACCOUNT

- **Operating Bank Account:** Corporate account for daily batch settlement.
- **T+1 Auto-Settlement:** Direct nodal account clearing with verified IFSC.
- **Webhook Diagnostic:** Ping button verifying gateway health.

COD RISK CONTROLS

- **COD Toggle:** Master enable/disable switch.
- **Product Restriction:** Strictly prohibited for prescription drugs (Rx); limited to pet food.
- **Maximum COD Value Cap:** Maximum threshold (₹2,500) per order.

11. Delivery & 60-Minute Express Settings

`#delivery-settings`

Hyperlocal service radii, micro-hub dispatch SLAs, carrier routing, and pharmaceutical cold-chain enforcement.

60-MINUTE HYPERLOCAL DELIVERY RULES

- **Express Service Radius:** Configurable boundary (default 5.0 km) from the nearest dark store.
- **Driver Allocation Timeout:** Auto-escalates if a rider is not confirmed within 5 minutes of order packing.
- **Free Express Delivery Cap:** Minimum order value (e.g., ₹499) to waive standard delivery fees.
- **Standard Delivery Fee:** Baseline flat fee applied to below-threshold carts.

PHARMACEUTICAL COLD-CHAIN MANDATE

- **Mandatory Cold-Box Policy:** Enforced for all temperature-sensitive vaccines, insulins, and biological serums.
- **Temperature Window:** Strictly enforced between +2°C and +8°C.
- **Dispatch Verification:** Requires physical data logger confirmation before rider handover.
- **Fleet Management:** Primary dedicated electric vehicle (EV) fleet + on-demand 3PL overflow.

12. API Credentials & External Integrations

`#api-integrations`

Cryptographic API secret management, rate limiters, and microservice bridges connecting ERPs, pathology labs, and WhatsApp.

PRODUCTION API CREDENTIALS

- **Live Secret Key:** Masked cryptographic token with interactive `Reveal` and `Copy` controls.
- **HMAC Signature:** Cryptographically signed requests preventing man-in-the-middle tampering.
- **Rate Limiting:** Enforced at 1,000 requests per minute per authenticated client ID.
- **Key Regeneration:** "+ Generate API Key" modal to instantly roll expired credentials.

ACTIVE MICROSERVICE CONNECTORS

- **Tally ERP Prime Bridge:** Real-time financial voucher and invoice sync.
- **Gupshup WhatsApp Cloud API:** High-throughput template notification dispatcher.
- **Central Pathology LIMS Connector:** Ingests external blood and biopsy lab reports.
- **Shiprocket / Shadowfax Logistics:** Live rider GPS tracking and reverse pickup dispatch.

13. Security & Zero-Trust Governance

`#security`

Zero-trust authentication, mandatory 2FA, enterprise Single Sign-On, subnet IP whitelisting, and session forensics.

AUTHENTICATION SECURITY

- **Mandatory 2FA:** Enforced across doctors, admins, and pharmacists.
- **Enterprise SSO:** Google Workspace and Microsoft Azure AD integration.
- **Session Inactivity Timeout:** Auto-logs out dormant sessions after 15 minutes.

CLINICAL SUBNET IP WHITELIST

- **Authorized IP Ranges:** Configurable CIDR blocks (e.g., `192.168.1.0/24`).
- **Confidentiality Rule:** Direct access to patient medical histories restricted to verified clinic subnets.

ACTIVE SESSION MANAGEMENT

- **Live Device Fingerprinting:** Tracks browser, IP address, and geographic location.
- **Current Session Tag:** Identifies active browser.
- **Remote Revocation:** Immediate `Revoke` button to terminate compromised sessions.

14. Disaster Recovery & Automated Snapshots

`#backup-recovery`

Continuous database WAL archiving, cross-region replication, Point-In-Time Recovery (PITR), and snapshot verification.

BUSINESS RESILIENCY KPIS

- **RPO (Recovery Point Objective):** < 5 Minutes maximum acceptable data loss.
- **RTO (Recovery Time Objective):** < 15 Minutes maximum system restoration downtime.
- **PITR Retention Window:** 30 Days continuous point-in-time recovery rollbacks.

CROSS-REGION RESILIENCY

- **Primary Region:** AWS / GCP Mumbai (`ap-south-1`) master cluster.
- **Secondary Region:** Hyderabad (`ap-south-2`) live synchronous read replica.
- **Automated Failover:** Health probes trigger DNS switchover on primary cluster failure.

SNAPSHOT AUDIT LEDGER

- **Manual Snapshot Trigger:** `Create Manual Snapshot` with immediate status.
- **Cryptographic Verification:** SHA-256 checksum integrity verification.
- **Snapshot Retention:** Secure S3 bucket with immutable Object Lock protection.